Combining （7)、(9) gives

$$\sin^{\alpha}\varphi_1\cos^{\alpha-2}\varphi_1=\sin^{\alpha}\varphi_2\cos^{\alpha-2}\varphi_2\text{。}\qquad(10)$$

If $\varphi_1\neq\varphi_2$, the following equation must have two distinct roots

$$\sin^{\alpha}\varphi\cos^{\alpha-2}\varphi-C=0\qquad(11)$$

in $\left(0,\frac{\pi}{2}\right)$, for a suitable constant C. The function $\sin^{\alpha}\varphi\cos^{\alpha-2}\varphi$ must therefore be nonmonotonic on $\left(0,\frac{\pi}{2}\right)$; $\sin^{\alpha}\varphi$ and $\cos^{\alpha-2}\varphi$ cannot both be increasing or both decreasing. As φ grows, if $\sin^{\alpha}\varphi$ rises, $\cos^{\alpha-2}\varphi$ must fall; if $\sin^{\alpha}\varphi$ falls, $\cos^{\alpha-2}\varphi$ must rise. Thus α and $\alpha-2$ have the same sign, giving

$$\alpha<0\qquad(12)$$

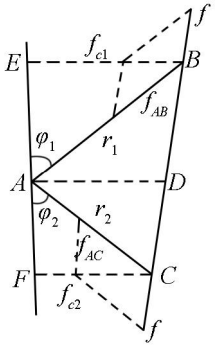
$$a>2\text{。}\qquad(13)$$

For $a<0$, $\sin^{-\alpha}\varphi\cos^{2-\alpha}\varphi$ is zero at $\varphi=0$ and $\frac{\pi}{2}$. Hence $\sin^{-\alpha}\cos^{2-\alpha}\varphi$ on $\left(0,\frac{\pi}{2}\right)$ has an extremum there. Thus （11) has at least two distinct roots for a suitable C. Likewise for $a>2$. Satisfying either of （12)、(13) is necessary and sufficient.

Scoring (62)(10)(1) ~ (4) each ; ; ; and the argument below them: 4 points in total.

Method 2:

In the figure, B、C repel with f; attractions from A are f_{AB}、f_{AC}. Their centripetal forces are f_{C1}、f_{C2}; distances to A are r_1、r_2. Similar triangles give



$$\frac{f_{AB}}{r_1}=\frac{f_{C1}}{AB}=\frac{f}{BD},\qquad(1a)$$

$$\frac{f_{AC}}{r_2}=\frac{f_{C2}}{AD}=\frac{f}{CD}\text{。}\qquad(2a)$$

Dividing the two equations gives

$$\frac{f_{AB}}{f_{AC}}\frac{r_2}{r_1}=\frac{f_{C1}}{f_{C2}}=\frac{CD}{BD}\qquad(3a)$$

The given conditions imply

$$\frac{f_{AB}}{f_{AC}}=\left(\frac{r_1}{r_2}\right)^{\alpha},\qquad(4a)$$

$$\frac{f_{C1}}{f_{C2}}=\frac{EB}{FC}=\frac{r_1\sin\varphi_1}{r_2\sin\varphi_2},\qquad(5a)$$